

Final Oral Exam Requirement – ET4278 2026

Description

The final assignment is based on the study and simulation of the **architecture** of a state-of-the-art sigma-delta (SD) modulator. Students will learn how to analyze a modern SD modulator, with an emphasis on understanding the operation of the chosen SD modulator architecture and its main design considerations.

To complete the final exam, each student should follow the following steps:

1. On a first-come-first-served basis, choose one of the papers from the list.
2. Study the paper carefully, with an emphasis on understanding the operation of the chosen SD modulator architecture and its main design considerations.
3. Develop a Matlab model that simulates the SD modulator described in the chosen paper. The model should also include the decimation filter. It should be sophisticated enough to substantiate the key claims of the paper e.g. by reproducing the simulation results shown in the paper. Note: For many of the papers, more details, such as loop filter coefficients, can be found from the associated MSc or PhD theses, which can often be found on-line.
4. Summarize your findings in a presentation. See below for a complete list of deliverables.
5. Bring your presentation and a laptop to the oral exam. During the oral exam, the simulated results will be discussed and evaluated. Moreover, more general questions on the topics covered in the course will be asked. The grade for the project represents 50% of the final grade, while the grade for the answers to general questions represents 20% of the final grade. The other 30% of the final grade is based on the results of the weekly assignments.

To complete the final exam, each student should follow the following steps in order and document their results in a presentation:

1. Identify the issues addressed by the assigned paper and identify the main techniques used to address these issues. (5 points)
2. Simulate the SD architecture described in the paper using a Matlab script. The simulation should reproduce the main results presented in the paper. Include your Matlab script and the simulation results in a **short** electronic report. Show both the results shown in the paper and your simulated results (15 points)
 - Enable and disable the proposed techniques in the paper. Quantify and compare the results.
 - Scale the voltage levels in the modulator to match those reported in the paper.
 - Add thermal noise and scale the output spectrum to match the results shown in

the paper. If applicable, plot the output spectrum before and after decimation.

3. Introduce the following circuit non-idealities in the Matlab script and show how they affect modulator performance. (5 points)
 - Finite amplifier gain & BW
 - DAC mismatch
 - Excess Loop Delay (ELD)
 - Clock jitter
 - Non-linearity in the integrator
4. Describe the circuit implementation of the various blocks. (5 points)
5. Identify the main techniques used in the SD modulator architecture. Discuss their weaknesses and strengths in the light of the current state of the art. (5 points)
6. Compare the adopted techniques to those used in modern SD modulators. The comparison should include (but is certainly not limited to) some of the references cited in the paper. Extend the comparison table given in the paper with the specifications of other notable designs (5 points)
7. Propose improvements to the SD modulator (5 points) and verify your proposed improvements with Matlab simulations. (5 points)

Final deliverables (must be submitted by email AT LEAST 24 hours before the scheduled oral exam. Failure to do this will result in cancellation of the exam):

1. Presentation file: Each question can be answered by using multiple slides but must be numbered properly. Your answers/conclusions should be included on each slide.
2. Matlab codes should also be submitted but separately from the slides.
3. TA's email address: yliu73@tudelft.nl

Good luck with the exam.